On Undecidable Propositions in Nondeterministic Languages and Related Systems

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Abstract

The definition of **NP** requires, for each member language L, a polynomial-time checking relation R and a constant k such that $w \in L \iff \exists y\,(|y| \leq |w|^k \wedge R(w, y))$. We show that this biconditional constructs, for each member language, exactly Hilbert's triple: a sound, complete, decidable proof system in which truth-in-L and bounded provability coincide by definitional fiat. We then demonstrate that the polynomial-time restriction on R does not shelter this construction from Gödel's reach, because Gödel's own proof-checking relation is already polynomial-time and fits inside the definition as a literal instance. The class **NP**, taken as a totality over all polynomial-time R, therefore contains languages for which the biconditional asserts a property that Gödel's First Incompleteness Theorem prohibits.

1 Introduction

The **P** vs **NP** problem has resisted resolution for over half a century, and barrier results have shown it cannot be settled by relativization [3], natural proofs [8], or algebrization [1]. This paper argues that the resistance is not a feature of the problem's difficulty but of its formulation: the definition of **NP** contains a semantic condition that is unsatisfiable for a class of its own instances.

The argument has three steps. Section 2 reproduces the standard definitions of **P** and **NP** and identifies the structural content of the biconditional in the **NP** definition. Section 3 establishes that Gödel's proof-verification relation is polynomial-time, confirming it is a valid instantiation of the checking relation R. Section 4 proves the main result: that the **NP** definition, so instantiated, reproduces the structure of Hilbert's Program—and is therefore subject to the same impossibility.

2 The Definition of NP and Its Structural Content

We follow the formulation of the Clay Millennium Prize statement [4].

A *checking relation* is a binary relation $R \subseteq \Sigma^* \times \Sigma_1^*$. We say R is *polynomial-time* iff the language $L_R = \{w\#y \mid R(w, y)\}$ is in **P**. A language L over Σ is in **NP** iff there exists $k \in \mathbb{N}$ and a polynomial-time checking relation R such that for all $w \in \Sigma^*$,

$$w \in L \iff \exists y \left(|y| \leq |w|^k \text{ and } R(w, y) \right). \tag{1}$$

2.1 What the Biconditional Demands

The biconditional (1) asserts three properties simultaneously:

1. **Soundness** ($\Leftarrow$): if R accepts the pair (w, y), then $w \in L$. No spurious witnesses exist.

2. **Completeness** ($\Rightarrow$): if $w \in L$, then some witness y of length at most $|w|^k$ exists. No members of L lack bounded certificates.

3. **Decidability**: R is total and polynomial-time. Verification always halts.

This triple—a verification procedure that is sound, complete over its domain, and mechanically decidable—is precisely the structure that Hilbert demanded for all of mathematics. Hilbert's Program required a formal system that would decide every mathematical statement by finitary means: sound (no false theorems), complete (every truth is provable), and effectively checkable (proofs are mechanically verifiable). The definition of **NP** assumes this triple into existence for every language in the class.

3 Gödel's Proof-Checking Relation is Polynomial-Time

Gödel's proof-checking relation $\mathit{ProofOf}(\pi, \varphi)$ (Definition 45 of [6]) determines whether π is a valid derivation whose last formula is φ. Under the natural string encoding—where a proof is represented as a sequence of formula-strings rather than a single integer via prime products—verification proceeds as follows:

1. **Linear scan.** The algorithm iterates through the k lines of the proof.

2. **Per-line check.** For each line, the algorithm pattern-matches against finitely many axiom schemata and searches preceding lines (at most $O(k^2)$ pairs) for a match via modus ponens or generalization.

3. **Total cost.** Each pattern-match is linear in the formula length m. The total verification cost is $O(k^2 \cdot m)$, polynomial in the input.

This is well known—proof verification is the standard textbook example [2] of an **NP** witness-checking procedure. The point that matters here is its consequence: Gödel's *ProofOf* is a *valid instantiation* of the checking relation R in the definition of **NP**. The polynomial-time restriction on R does not exclude Gödel's system; it includes it.

Two properties of *ProofOf* are essential:

- It is *decidable*: every quantifier in Definitions 1–45 is bounded; the algorithm always halts with TRUE or FALSE.

- It is *distinct from Bew*: the predicate $\mathit{Bew}(\varphi) \equiv \exists\pi\,[\mathit{ProofOf}(\pi, \varphi)]$ adds an *unbounded* existential quantifier, making it recursively enumerable but not decidable. This is the boundary at which decidability fails.

The **NP** definition attempts to walk a middle path: it uses an *existential quantifier*, like *Bew*, but *bounds* it, like *ProofOf*. This is *Bew* with a leash.

4 The Equivalence

We now prove the main claim.

4.1 Instantiation

Let T be any consistent, recursively axiomatizable theory interpreting Robinson arithmetic Q (e.g., Peano Arithmetic). Define:

- $R = \mathit{ProofOf}_T$, the proof-checking relation for T. By Section 3, R is polynomial-time.
- For each $k \in \mathbb{N}$, define the language

$$L_k = \{\varphi \in \Sigma^* : \exists \pi \, (|\pi| \leq |\varphi|^k \ \wedge \ \mathit{ProofOf}_T(\pi, \varphi))\}.$$

Each L_k is in **NP** by construction: the checking relation is polynomial-time, the witness bound is $|\varphi|^k$, and the biconditional (1) holds—trivially, because L_k is *defined* as the right-hand side.

4.2 What the Biconditional Asserts for This Instance

For L_k, the biconditional reads:

$$\varphi \in L_k \iff \exists \pi \, (|\pi| \leq |\varphi|^k \ \wedge \ \mathit{ProofOf}_T(\pi, \varphi)).$$

The completeness direction ($\Rightarrow$) asserts: *every member of L_k has a proof in T of length at most $|\varphi|^k$*. This is true by definition—L_k was defined to contain only such sentences. But consider what L_k *is*: it is the set of theorems of T that have short proofs. It is a proper subset of $\mathrm{Thm}(T)$, which is itself a proper subset of the true sentences of arithmetic (by Gödel's First Incompleteness Theorem).

The biconditional is satisfied, but only because L_k has been *carved* to exclude:

1. Theorems of T whose shortest proof exceeds $|\varphi|^k$ (these exist by proof-complexity lower bounds [7]).
2. True sentences that T cannot prove at all (the Gödel sentence G_T).

4.3 The Defect at the Class Level

The definition of **NP** does not merely assert that L_k exists for one value of k. The class is defined as:

$$\mathbf{NP} = \bigcup_{k \in \mathbb{N}} \bigcup_{R \in \text{poly-time}} \{L : \forall w \, (w \in L \iff \exists y \, (|y| \leq |w|^k \wedge R(w, y)))\}.$$

This quantifies over *all* polynomial-time checking relations R and *all* constants k. The class therefore necessarily includes, as instances, every L_k constructed from every sufficiently strong theory T.

Now apply Gödel's theorem. For any fixed T:

1. By the First Incompleteness Theorem, there exists a sentence G_T such that $T \nvdash G_T$ and $T \nvdash \neg G_T$.
2. $G_T \notin L_k$ for any k, because G_T has no T-proof at all.
3. Therefore no single L_k (and no finite union of them) captures all theorems of T, let alone all truths.

One might respond: extend to a stronger theory $T' = T + G_T$. But by the argument of upward propagation, T' has its own independent sentence $G_{T'}$, constructed from T''s own proof predicate via the Fixed-Point Lemma. The diagonal construction regenerates at every level. There is no consistent, recursively axiomatizable theory whose theorems are all captured by *any* single **NP** language.

4.4 The Structural Identification with Hilbert's Program

The parallel is now exact:

Hilbert's Program	**The NP definition**
Formal system $\mathfrak{F}$	Checking relation R with bound k
Sound: $\mathfrak{F} \vdash \varphi \Rightarrow \varphi$ is true	$R(w, y) \Rightarrow w \in L$
Complete: φ true $\Rightarrow \mathfrak{F} \vdash \varphi$	$w \in L \Rightarrow \exists y\,(\lvert y\rvert \leq \lvert w\rvert^k \wedge R(w, y))$
Decidable by finitary means	R polynomial-time
Killed by Gödel's theorem	Killed by Gödel's theorem

Hilbert demanded a single formal system satisfying all three properties for all of mathematics. Gödel showed no such system exists. The definition of **NP** demands the same triple for each member language—and since the class quantifies over all polynomial-time R, it necessarily includes instances where R encodes proof verification for systems strong enough to trigger Gödel's theorem. For those instances, the biconditional asserts that bounded provability captures truth. It does not.

The polynomial bound $|y| \leq |w|^k$ was supposed to create immunity: restrict to phenomena with short certificates, and the Gödelian pathology—true sentences with no proof—cannot appear. But the immunity is illusory. The bound excludes individual Gödel sentences (whose proofs, if they existed, would be long). It does not exclude the *structure* that generates them. Gödel's proof-checker is polynomial-time. The Fixed-Point Lemma operates within any system that can represent its own proof predicate. The diagonal construction does not require super-polynomial resources—it requires only that the system can talk about itself, which any system interpreting Q can do.

5 Proof Complexity and the Point of Departure

The most developed formal connection between Gödel's incompleteness theorems and the **P** vs **NP** problem runs through propositional proof complexity, and in particular through the Cook–Reckhow framework [5] and the theory built on it by Krajíček [7]. This section traces that connection and identifies the precise point at which the present argument departs from it.

5.1 The Cook–Reckhow Theorem

Cook and Reckhow [5] defined a *propositional proof system* as a polynomial-time computable surjection $P : \{0,1\}^* \to \mathrm{TAUT}$. A proof system P is *p-bounded* if there exists $c \geq 1$ such that every tautology α has a P-proof π with $|\pi| \leq (|\alpha| + c)^c$. Their central result is a clean biconditional:

Theorem 5.1 (Cook–Reckhow [5])**.** *A p-bounded propositional proof system exists if and only if* $\mathbf{NP} = \mathbf{coNP}$.

The theorem translates a question about complexity classes into a question about the existence of a proof system in which every tautology has a short proof. Crucially, the theorem is agnostic: it is silent about which side holds.

5.2 The Quantitative Gödel Theorem

Krajíček [7] showed how Gödel's incompleteness theorems enter this framework. For any consistent, recursively axiomatizable theory T extending S^1_2, one defines a propositional proof system P_T that operates as extended resolution augmented by propositional translations of T's theorems. Krajíček observes that the provability predicate $\mathrm{Prf}_T(x, z)$ is polynomial-time decidable—and that this holds for all standard calculi, since "the verifying algorithm essentially needs only to decide repeatedly whether a string is a formula, or whether one string is a substitution instance of another." Standard first-order theories such as PA and ZFC, axiomatized by finitely many schemes, thus yield Cook–Reckhow proof systems whose proof-verification relations are polynomial-time.

Gödel's Second Incompleteness Theorem then bites. The consistency statement Con_T is a true Π^0_1-sentence, and its propositional translations $\|\mathrm{Con}_T\|^n$ are genuine tautologies. But since $T \nvdash \mathrm{Con}_T$, the system P_T cannot derive these tautologies from its axioms, and their P_T-proofs must grow super-polynomially. Krajíček makes this quantitative:

Theorem 5.2 (Friedman; Pudlák; see Krajíček [7], Theorem 21.3.1)**.** *Let $T \supseteq S^1_2$ be a finite, consistent theory. There exists $\epsilon > 0$ such that for all $m \geq 1$, $T \nvdash_{m^\epsilon} \mathrm{Con}_T(\underline{m})$.*

This is the migration channel: Gödel's incompleteness, which in first-order logic produces sentences that are true but *unprovable*, becomes in the propositional setting a *proof-length lower bound*. The incompleteness has not disappeared—it has moved from the binary boundary between provable and unprovable to the quantitative boundary between polynomial and super-polynomial proof length. The diagonal mechanism driving it is the same: the modified diagonal formula $\delta(\underline{m}) \equiv \neg \mathrm{Pr}^{\underline{m}}_T(\lceil \delta(\underline{m}) \rceil)$ is constructible in polynomial size, and Krajíček's Löb conditions $1'$–$4'$ carry the standard proof through with explicit proof-length accounting.

The consequence is that each proof system P_T generates consistency tautologies requiring a stronger system. A p-bounded proof system would be a fixed point—a system that efficiently proves its own consistency tautologies. Gödel's theorem says no consistent system is such a fixed point.

5.3 What the Framework Concludes, and Where This Paper Departs

Despite establishing this Gödelian structure, the standard conclusion within proof complexity is *methodological*, not foundational. As Krajíček writes: "unless there is an optimal proof system you cannot hope to prove that **NP** $\neq$ **coNP** by gradually proving super-polynomial lower bounds for stronger and stronger proof systems as that would be an infinite process." The Gödelian hierarchy is treated as a constraint on *proof strategies*: it tells us that no single proof system can be shown to be p-bounded by exhibiting it directly, because each candidate fails on its own consistency tautologies. But within the standard framework, this infinite regress is regarded as a feature of the *landscape of proof systems*, not as evidence that the class **NP** is itself defective.

The architectural reason for this restraint is a sharp distinction between *individual* **NP** languages and the *class* **NP** as a whole. For any particular **NP** language L—composites, 3-colorable graphs, Hamiltonian paths—the checking relation R is a concrete, fixed predicate, and the biconditional $w \in L \iff \exists y\,(|y| \leq |w|^k \wedge R(w, y))$ is simply a definition of what L is. There is no independent notion of "truth in L" that could diverge from bounded provability. Gödel's theorem does not bite individual languages unless the language encodes its own proof predicate, and most **NP** languages do not. The Cook–Reckhow framework therefore quantifies over all proof systems and asks whether *some* system is p-bounded; the fact that each particular system fails does not, within this framework, entail that no system can succeed.

The present paper does not operate within this framework. Rather than asking whether some proof system escapes the Gödelian hierarchy, it identifies the structural assumption that *generates* the hierarchy in the first place: the biconditional in the definition of **NP** itself, which asserts, for each member language L, that membership and bounded verifiability coincide. The class **NP**, taken as a totality, asserts that this biconditional works for *every* polynomial-time checking relation R and *every* bounding constant k. But the moment one quantifies over all such R, one necessarily includes checking relations expressive enough to encode proof verification for formal systems—since Gödel's proof-checker is polynomial-time, it fits inside the definition as a literal instance.

The distinction can be stated precisely. Krajíček's result says:

> For each proof system P, there exist tautologies whose P-proofs are super-polynomial. *(A constraint on proof strategies.)*

The present paper argues:

> For each sufficiently strong theory T, the **NP** language $L_k = \{\varphi : \exists\pi\,(|\pi| \leq |\varphi|^k \wedge \mathrm{Prf}_T(\pi, \varphi))\}$ satisfies its defining biconditional only by excluding—by construction—all sentences that lack bounded proofs, including the Gödel sentence G_T. The definition of **NP** presupposes that such exclusion is unproblematic. But the class quantifies over all polynomial-time R, and for those R that encode proof verification, the "excluded" sentences are precisely the ones whose existence Gödel's theorem guarantees. The biconditional does not fail; it is satisfied vacuously, by defining away the phenomena it cannot capture. *(A defect in the definition.)*

The impossibility, on this analysis, does not live in the Cook–Reckhow hierarchy of proof systems. It lives in the definition that the hierarchy presupposes.

5.4 The Conclusion

The definition of **NP** presupposes, for each member language, a property—the simultaneous satisfaction of soundness, completeness, and decidability—that Gödel's theorem proves cannot be simultaneously satisfied for any language whose checking relation encodes proof verification in a sufficiently strong theory. Since such languages are legitimate instances of the definition (the checking relation is polynomial-time, the biconditional holds by construction), the class **NP** contains instances whose defining condition is modeled on a structure already known to be impossible.

The question $\mathbf{P} = \mathbf{NP}$ asks whether verification can always be converted to efficient decision. But this question inherits the structural defect of its premise: it presupposes that the class **NP** is a coherent mathematical object, which requires that the biconditional (1) can be simultaneously satisfied across all its instances. For the instances that encode proof verification, it cannot.

References

[1] Scott Aaronson and Avi Wigderson. Algebrization: A new barrier in complexity theory. In *ACM Transactions on Computation Theory*, volume 1, pages 2:1–2:54, 2009.

[2] Sanjeev Arora and Boaz Barak. *Computational Complexity: A Modern Approach*. Cambridge University Press, 2009.

[3] Theodore Baker, John Gill, and Robert Solovay. Relativizations of the $\mathcal{P} = ? \mathcal{NP}$ question. *SIAM Journal on Computing*, 4(4):431–442, 1975.

[4] Stephen A. Cook. The P versus NP problem. In James Carlson, Arthur Jaffe, and Andrew Wiles, editors, *The Millennium Prize Problems*, pages 87–104. American Mathematical Society and Clay Mathematics Institute, 2006.

[5] Stephen A. Cook and Robert A. Reckhow. The relative efficiency of propositional proof systems. *Journal of Symbolic Logic*, 44(1):36–50, 1979.

[6] Kurt Gödel. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931. English translation in [9].

[7] Jan Krajíček. *Proof Complexity*, volume 170 of *Encyclopedia of Mathematics and its Applications*. Cambridge University Press, 2019.

[8] Alexander A. Razborov and Steven Rudich. Natural proofs. *Journal of Computer and System Sciences*, 55(1):24–35, 1997.

[9] Jean van Heijenoort, editor. *From Frege to Gödel: A Source Book in Mathematical Logic, 1879–1931*. Harvard University Press, 1967.